

Simulation-Based Inference for an Adaptive-Network SIR Model

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1 Introduction

[TODO: Model motivation, data description, why likelihood is intractable. Suggested length: ~1 page.]

2 Model and Data

[TODO: Formal model description: Erdős–Rényi network ($N = 200$, $p = 0.05$, 5 initially infected), three synchronous operations (Infection, Recovery, Rewiring), and why the adaptive rewiring makes the likelihood intractable. Reference ?. Suggested length: ~0.5 pages.]

The three unknown parameters have independent uniform priors: $\beta \sim \mathcal{U}(0.05, 0.50)$, $\gamma \sim \mathcal{U}(0.02, 0.20)$, $\rho \sim \mathcal{U}(0.0, 0.8)$, with prior standard deviations $\sigma_\beta^{\text{prior}} = 0.130$, $\sigma_\gamma^{\text{prior}} = 0.052$, $\sigma_\rho^{\text{prior}} = 0.231$. The observed dataset comprises $R = 40$ independent replicates, each providing an infection time series, a rewiring time series, and a final degree distribution (Figure 1).

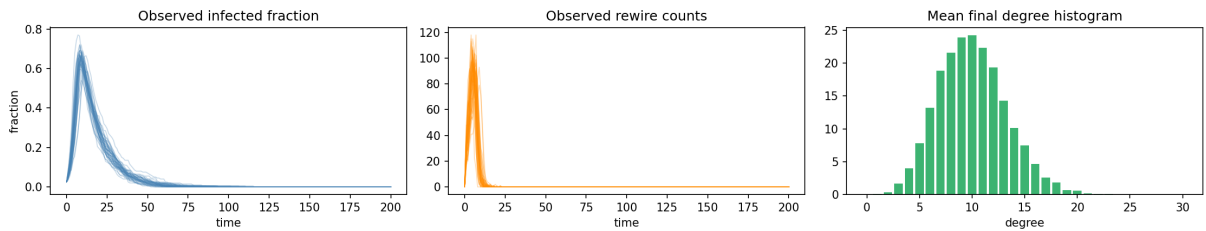


Figure 1: Observed data across 40 replicates: infection fraction (left), rewiring events (centre), final degree distribution (right).

3 Summary Statistics

Because acceptance probability in ABC falls exponentially with summary dimension (?), we compress the raw time series to a 15-dimensional *balanced* feature vector averaged across replicates: five infection summaries (peak fraction, epidemic duration, total infection, and their cross-replicate standard deviations), four rewiring summaries (total events, peak rate, and standard deviations), and six degree summaries (mean, std, entropy, quartiles of the final degree distribution).

3.1 Identifiability diagnostics

Projecting the $M = 10\,000$ prior-simulation summaries to 2D via PCA and colouring by parameter value (Figure 2) reveals clear gradients for γ , moderate structure for β , and weak but non-trivial structure for ρ . This confirms the central identifiability challenge: β and ρ both suppress the epidemic — via higher transmission rate being countered by faster recovery (γ) or network rewiring (ρ) — so infection-curve summaries alone cannot fully separate them. Rewiring summaries carry independent information about both β and ρ .

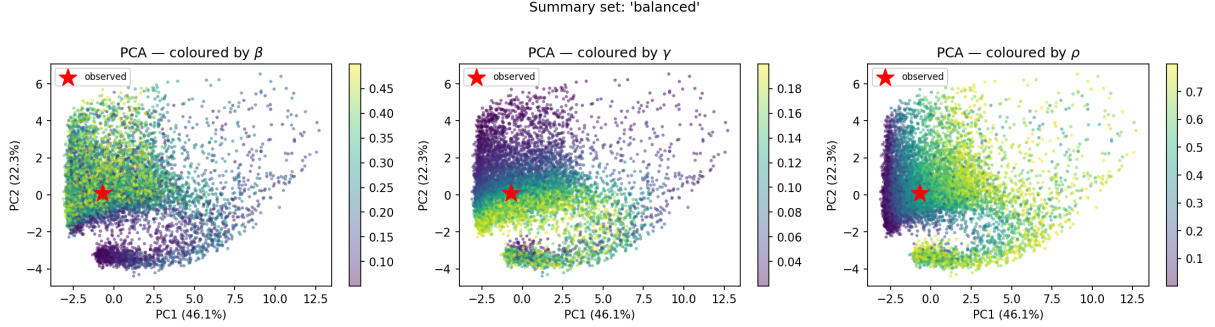


Figure 2: PCA diagnostic: prior simulations coloured by β , γ , ρ (left to right). Clear gradients for γ ; partial confounding visible in β – ρ panels.

3.2 Summary-set comparison

We evaluated four candidate sets at $q = 1\%$. Table 1 and Figure 3 report posterior std relative to prior std (lower = more informative).

Table 1: Relative posterior std (std / prior std) per summary set at $q = 1\%$. Bold marks the best set for each parameter.

Set	β	γ	ρ
infection_only	0.377	0.084	0.854
rewiring_only	0.216	0.804	0.173
degree_only	0.924	0.932	0.483
balanced	0.479	0.378	0.262

The results reveal a clear separation of information sources: γ is almost entirely encoded in the infection curve (*infection_only* reduces its posterior to 8% of the prior width), while both β and ρ are best identified from rewiring summaries (*rewiring_only* achieves relative widths of 0.22 and 0.17 respectively). The degree distribution alone carries little information about β or γ , but provides moderate constraint on ρ (0.48).

No single-source set identifies all three parameters well simultaneously. The *balanced* set is a deliberate compromise: it sacrifices some per-parameter sharpness in exchange for robustness across all three, and is used in all subsequent analyses.

4 Rejection ABC

We drew $M = 10\,000$ prior samples, computed the standardised Euclidean distance between each simulated and observed summary vector, and accepted draws below the 1% empirical quantile (?), giving $\varepsilon = 1.2852$ and $n_{\text{acc}} = 100$. Figure 4 shows the marginal posteriors; Table 2 gives the numerical summaries.

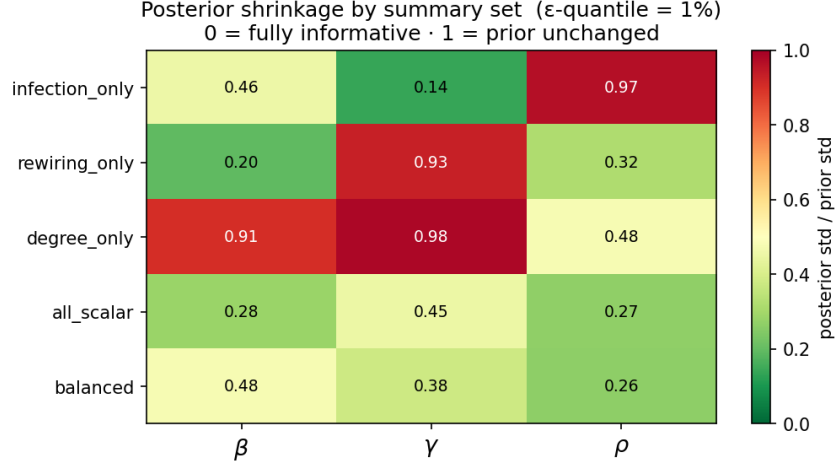


Figure 3: Posterior std (relative to prior) per summary set and parameter at $q = 1\%$. Lower is more informative.

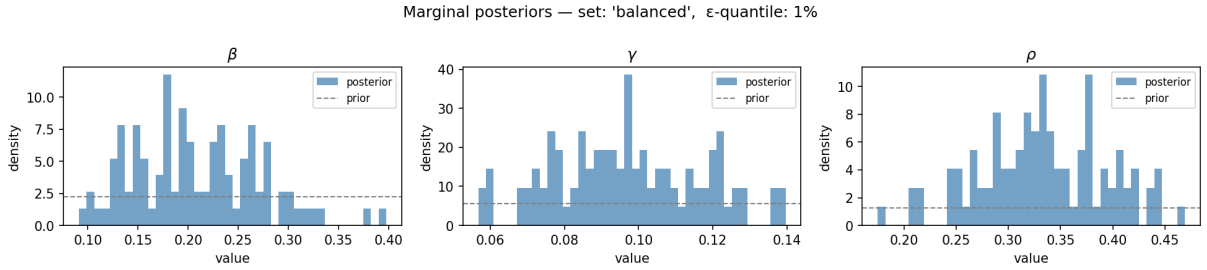


Figure 4: Rejection ABC marginal posteriors (blue) vs. uniform prior (dashed), *balanced* set, $\epsilon = 1.2852$, $n_{\text{acc}} = 100$.

γ is best-identified (directly controls epidemic duration), β moderately so, and ρ is hardest to constrain — consistent with the PCA diagnostic and the negative (β, ρ) posterior correlation driven by their shared epidemic-suppression effect.

5 Advanced Methods

5.1 Semi-Automatic ABC (?)

? show that the optimal ABC summaries are the posterior means $\mathbb{E}[\theta \mid x]$, which can be approximated by fitting Ridge regression from the balanced features to each parameter on the pilot draws. In-sample R^2 values confirm strong predictability: $R^2_{\beta} = 0.812$, $R^2_{\gamma} = 0.857$, $R^2_{\rho} = 0.979$. Running rejection ABC in the resulting 3-dimensional projected space ($\epsilon = 0.520$, $n_{\text{acc}} = 100$) substantially tightens the posterior for β and provides moderate gains for γ and ρ (Figure 5, Table 3).

5.2 LASSO-Based Summary Selection (?)

Fitting LassoCV from summaries to each parameter on the pilot simulations retained 14/15 features for β ($\alpha = 6 \times 10^{-4}$) and all 15 for γ and ρ ($\alpha = 8 \times 10^{-4}$). No feature is entirely redundant, so the LASSO-selected set is effectively identical to the balanced set and yields the same posterior (Table 3).

Table 2: Rejection ABC posterior ($q = 1\%$, *balanced*).

Param	Mean	Std	25%	Median	75%
β	0.208	0.063	0.159	0.201	0.254
γ	0.097	0.020	0.083	0.096	0.112
ρ	0.334	0.061	0.291	0.332	0.378

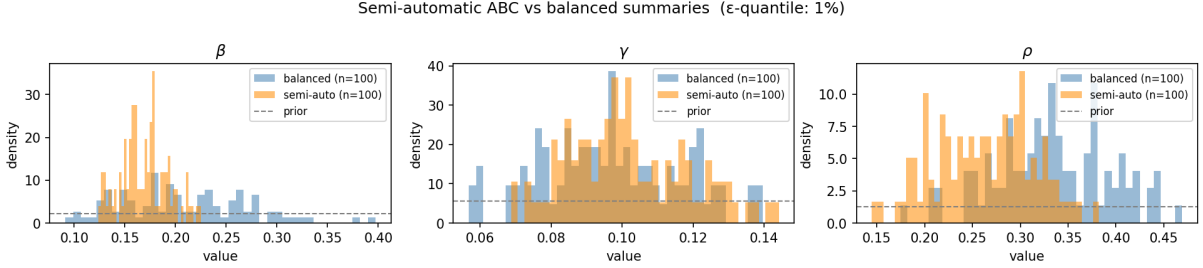


Figure 5: Balanced rejection ABC (blue) vs. semi-automatic ABC (orange), $q = 1\%$.

5.3 Regression Adjustment (?)

After accepting draws at a loose tolerance ($q = 5\%$, $\varepsilon = 1.706$, $n_{\text{acc}} = 500$), each accepted $\theta^{(i)}$ is corrected for the residual gap between its simulated summary $s^{(i)}$ and s_{obs} :

$$\tilde{\theta}^{(i)} = \theta^{(i)} - \hat{B}^\top (s^{(i)} - s_{\text{obs}}),$$

where $\hat{B}$ is estimated by weighted least squares with Epanechnikov kernel weights (?). Figure 6 and Table 4 show that adjustment dramatically sharpens the posterior (σ_β : $0.099 \rightarrow 0.024$, σ_γ : $0.033 \rightarrow 0.007$, σ_ρ : $0.087 \rightarrow 0.008$). Crucially, the adjusted posterior is nearly tolerance-invariant for $q \lesssim 10\%$ (Figure 7), whereas the raw posterior degrades rapidly at loose tolerances.

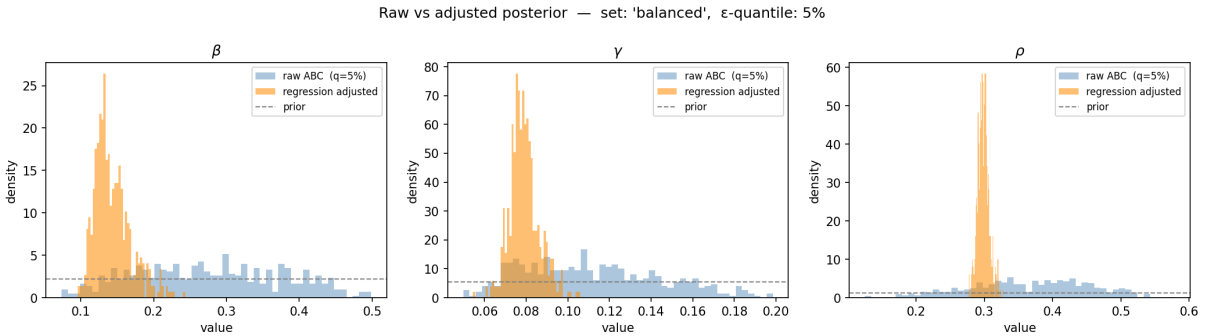


Figure 6: Raw ABC (blue) vs. regression-adjusted (orange) posteriors, $q = 5\%$, *balanced* set.

5.4 Synthetic Likelihood (?)

At each MCMC proposal θ , we simulate $N_{\text{sim}} = 50$ datasets ($R_{\text{sim}} = 40$ replicates each) and approximate the intractable likelihood as a multivariate Gaussian fitted to the N_{sim} summary vectors, using Ledoit–Wolf shrinkage for the covariance (?):

$$\log \hat{p}_{\text{SL}}(s_{\text{obs}} | \theta) = \log \mathcal{N}(s_{\text{obs}}; \hat{\mu}(\theta), \hat{\Sigma}(\theta)).$$

This is embedded in a Metropolis–Hastings random-walk MCMC ($N_{\text{steps}} = 2000$, burn-in = 500, step sizes $\sigma = (0.04, 0.015, 0.06)$).

Table 3: Posterior std and relative contraction (std/prior std) for rejection ABC variants at $q = 1\%$.

Method	β		γ		ρ	
	Std	Rel.	Std	Rel.	Std	Rel.
Prior	0.130	1.00	0.052	1.00	0.231	1.00
Balanced (15d)	0.062	0.48	0.020	0.38	0.061	0.26
LASSO (15d)	0.062	0.48	0.020	0.38	0.061	0.26
Semi-auto (3d)	0.023	0.18	0.017	0.33	0.051	0.22

Table 4: Raw vs. regression-adjusted posteriors ($q = 5\%$).

Method	Param	Mean	Std	5%	Median	95%
Raw ($q = 5\%$)	β	0.276	0.099	0.127	0.272	0.439
	γ	0.111	0.033	0.066	0.108	0.169
	ρ	0.367	0.087	0.216	0.373	0.504
Adjusted	β	0.143	0.024	0.112	0.138	0.191
	γ	0.079	0.007	0.069	0.078	0.092
	ρ	0.299	0.008	0.286	0.298	0.313

The acceptance rate was $\approx 1\text{--}3\%$ (target: 20–40%), resulting in very low effective sample sizes ($\text{ESS} \approx 4\text{--}5$ from 1 501 post-burn-in samples), as visible in the trace and autocorrelation plots (Figure 8). Despite poor mixing, the chain clusters consistently around $(\hat{\beta}, \hat{\gamma}, \hat{\rho}) \approx (0.152, 0.080, 0.302)$, in agreement with the regression-adjusted medians. The reported posterior standard deviations (Table 5) are therefore underestimates of the true posterior width. Remedies include reducing step sizes, using an adaptive proposal, or switching to a gradient-based sampler.

Table 5: Synthetic likelihood MCMC posterior (1 501 post-burn-in samples). Low ESS: std values are lower bounds on true posterior width.

Param	Mean	Std	5%	Median	95%	ESS
β	0.152	0.003	0.149	0.152	0.156	4
γ	0.080	0.002	0.077	0.082	0.082	5
ρ	0.302	0.007	0.295	0.300	0.314	4

6 Comparison

Table 6 summarises all methods. Point estimates are consistent across advanced methods ($\hat{\beta} \approx 0.14\text{--}0.21$, $\hat{\gamma} \approx 0.08\text{--}0.11$, $\hat{\rho} \approx 0.30$). Regression adjustment gives the most reliable uncertainty quantification: large accepted set ($n = 500$), tolerance-robust, and verifiable via sensitivity analysis. The SL chain provides a useful point estimate but requires better step-size tuning before its uncertainty intervals can be trusted.

[TODO: If additional methods (ABC-MCMC, SMC-ABC) were implemented, add a subsection here.]

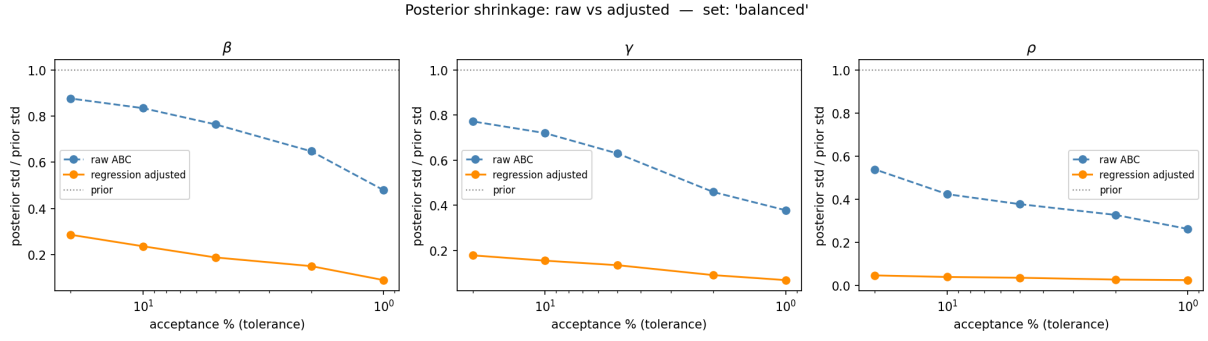


Figure 7: Posterior std (relative to prior) vs. tolerance q for raw (blue) and adjusted (orange) ABC. Adjustment is approximately tolerance-invariant for $q \lesssim 10\%$.

Table 6: Posterior standard deviations across all methods. [†]SL chain not converged; std values are lower bounds.

Method	σ_β	σ_γ	σ_ρ	Notes
Prior	0.130	0.052	0.231	—
Rej. ABC ($q = 1\%$)	0.062	0.020	0.061	$n = 100$
LASSO ($q = 1\%$)	0.062	0.020	0.061	All 15 features kept
Semi-auto ($q = 1\%$)	0.023	0.017	0.051	3-d projection
Reg. adj. ($q = 5\%$)	0.024	0.007	0.008	$n = 500$, robust
Synth. lik. [†]	0.003	0.002	0.007	ESS ≈ 4

7 Conclusion

[TODO: Summarise: (1) identifiability findings; (2) how each method improves on basic rejection ABC; (3) SL mixing limitations and future work. Suggested length: ~ 0.5 pages.]

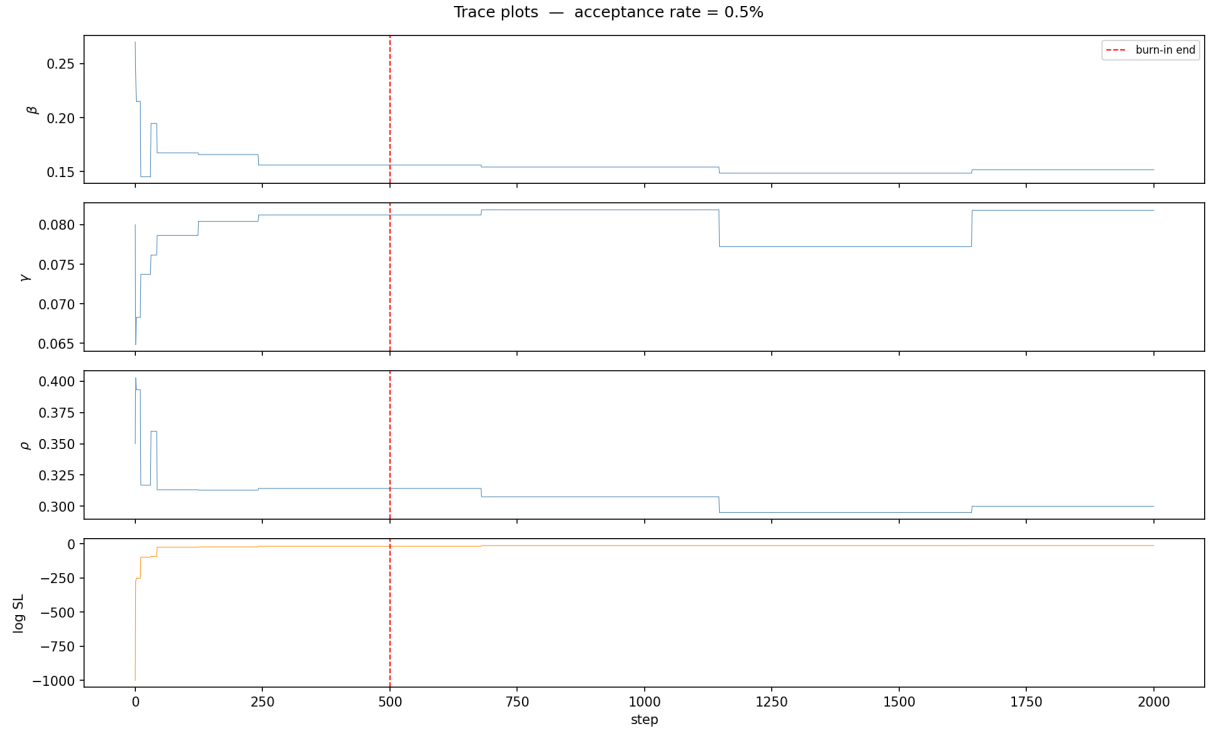


Figure 8: MCMC trace plots. The chain shows prolonged stagnation, reflecting the low acceptance rate. Autocorrelation remains near 1.0 throughout, confirming $ESS \approx 4-5$.

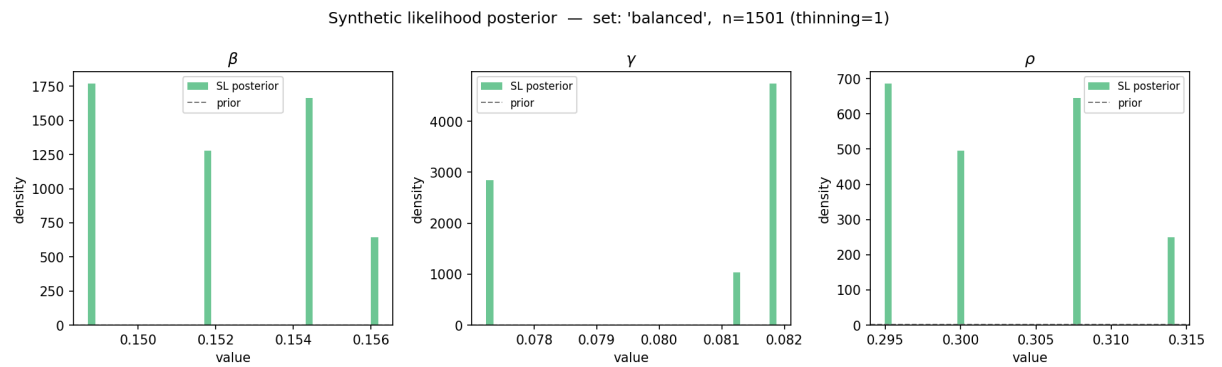


Figure 9: Synthetic likelihood MCMC posterior (green) vs. prior (dashed). Narrow widths reflect chain stagnation rather than true posterior precision.